Floe Foxon *Scientist·Analyst·Programmer*

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EXPERIENCE

Researcher	The Center for Scientific Integrity (Medical Evidence Project)	2026–Now
Scientist and Data Analyst	Pinney Associates	2024–2026
Statistician		2020–2024
Research Scholar	Sanford Research	2019
Python Programming Tutor	CodeX (Enactus Nottingham)	2018

EDUCATION

GCert Artificial Intelligence and Machine Learning Courses: AI in Astrophysics Machine Learning AI in Healthcare Deep Learning	University of Texas at Austin	2026–2027
MSc Data Science (Statistics) Courses: Programming for Data Science Statistical Computing Bayesian Statistics Linear Modelling	University of Leeds	2024–2025
BSc (Hons) Physics with Astronomy Courses: Computing for Physical Science Mathematics for Physics Scientific Computing	University of Nottingham	2017–2020
General Certificate of Education Advanced Level Subjects: Physics (Astrophysics) Computer Science Mathematics	Bishop Vesey's Grammar School	2015–2017

PROFESSIONAL DEVELOPMENT

Course: Simulation Modeling of Tobacco Use, Health Effects and Policy Impacts	University of Michigan	2026
Human Research Training Certificate	CITI (MIT Affiliates)	2026
Wolfram Summer Research Institute, Physics and Foundational Science Track	Wolfram Research, Inc.	2025
Data-Driven Astronomy Certificate	University of Sydney	2024
Professional Certificate Data Science	IBM	2020

SKILLS

Programming	Python (scikit-learn, PyTorch, TensorFlow, pandas, Matplotlib), Jupyter Notebook, R, SQL, SAS, MATLAB, Java, Wolfram
Software	GitHub, Microsoft Office 365/SharePoint, Google Workspace, Linux, Windows, Tableau, Mathematica

SELECTED MEDIA APPEARANCES

Science ◇ New Scientist ◇ Retraction Watch ◇ Popular Mechanics ◇ Photonics Focus

SELECTED WORKS

Foxon F, Wang D, Selya AS, Henningfield JE. 2026. The Need to Computationally Model and Simulate the Unintended Public Health and Economic Impacts of Kratom Regulation. Harm Reduction Journal. Under Review.

Foxon F. 2026. Computational Cryptids: Individual-Based Modelling and Simulation of Marine Animal Discovery with Implications for Undiscovered Species. Oral presentation at the Folk Zoology Conference 2026. Virtual.

Foxon F, Sembower MA, Shiffman S. 2026. Temporal Generalized Linear Mixed Models for Monitoring National Health Behavior in the United States Using Subnational Observational Data. Epidemiology and Health. 48:e2026035.

Foxon F. 2026. Modelling Extreme Precipitation in Aotearoa (New Zealand) from 1951–2025 with Bootstrapping, Negative Binomial Regression, and Bayesian Additive Regression Trees. Poster presentation at the Exeter Climate Forum 2026. Exeter, England.

Foxon F. 2026. Statistical Tests for Randomness on a Typewritten Key Stream Extracted With Computer Vision and Classified With a Convolutional Neural Network. Proceedings of the 9th International Conference on Historical Cryptology HistoCrypt 2026. NEALT Proceedings Series Number 61:224-229.

Foxon F, Niaura R. 2026. Logistic Regression Is Not Enough: The Need for Bayesian Nonparametric Modelling for Causal Inference Using Observational Data, Exemplified by the 'Gateway' Effect. Nicotine & Tobacco Research - Special Issue on Novel Methodologies in Nicotine, Tobacco, and Substance Use Epidemiology. Under revisions.

Foxon F. 2025. Mining Pockets of Computational Reducibility with AI: Transformer Models of Graph Rewriting Systems. Wolfram Community, Staff Picks.

Foxon F. 2025. Machine Learning for Text Classification in Classical Cryptography. Proceedings of the 8th International Conference on Historical Cryptology HistoCrypt 2025. NEALT Proceedings Series 58:60-64.

Foxon F. 2025. Solar Cycles: Can They Be Predicted? Research Notes of the American Astronomical Society. 9(2):40. Previously presented as a poster at the 245th meeting of the American Astronomical Society (AAS), National Harbor, Maryland: Forecasting the Amplitude of Solar Cycles. Bulletin of the AAS. 57(2). Presentation #105.01.

Foxon F. 2024. Artificial Neural Network for Hoax Cryptogram Identification. Proceedings of the 7th International Conference on Historical Cryptology HistoCrypt 2024. NEALT Proceedings Series Number 53:86-90.

Foxon F. 2023. Can Bayesian Statistics Be Used to Analyze Phenomena in Folk Zoology? Journal of Scientific Exploration. 37(3):570–571.

Foxon F. 2021. Evaluating Modern System Dynamics Software for Use in Astrophysical Simulations. Astronomy and Computing. 36:100486.

Foxon F. 2021. Ammonoid Taxonomy with Supervised and Unsupervised Machine Learning Algorithms. PCI Paleontology.

AWARDS

Distinction Award	University of Leeds	2026
Leeds Partnership Awards Nominee (×2), Shortlisted	University of Leeds	2026
Chambliss Student Poster Competition Finalist	American Astronomical Society	2025
First Class Honours Award	University of Nottingham	2020
First Year Scholarship Award for Academic Achievement	University of Nottingham	2018

SERVICES

Chair	Folk Zoology Conference 2024 (Virtual); Folk Zoology Conference 2026 (Virtual)
Ambassador	University of Nottingham (2018–2019); Students for Sensible Drug Policy (2026)
Peer Reviewer	Scientific Reports; PeerJ; American Journal of Health Behavior; Drug and Alcohol Dependence; Harm Reduction Journal; Internal and Emergency Medicine; Contributions to Tobacco & Nicotine Research; Addictive Behaviors; SSM - Qualitative Research in Health; Nicotine & Tobacco Research

RESEARCH INTERESTS

I am interested in all areas of science and mathematics, especially applications of machine learning and modelling/simulation techniques to novel data. My personal goal is to solve complicated problems, and I am passionate about working on innovative scientific and technical projects that meet cost and time budgets.